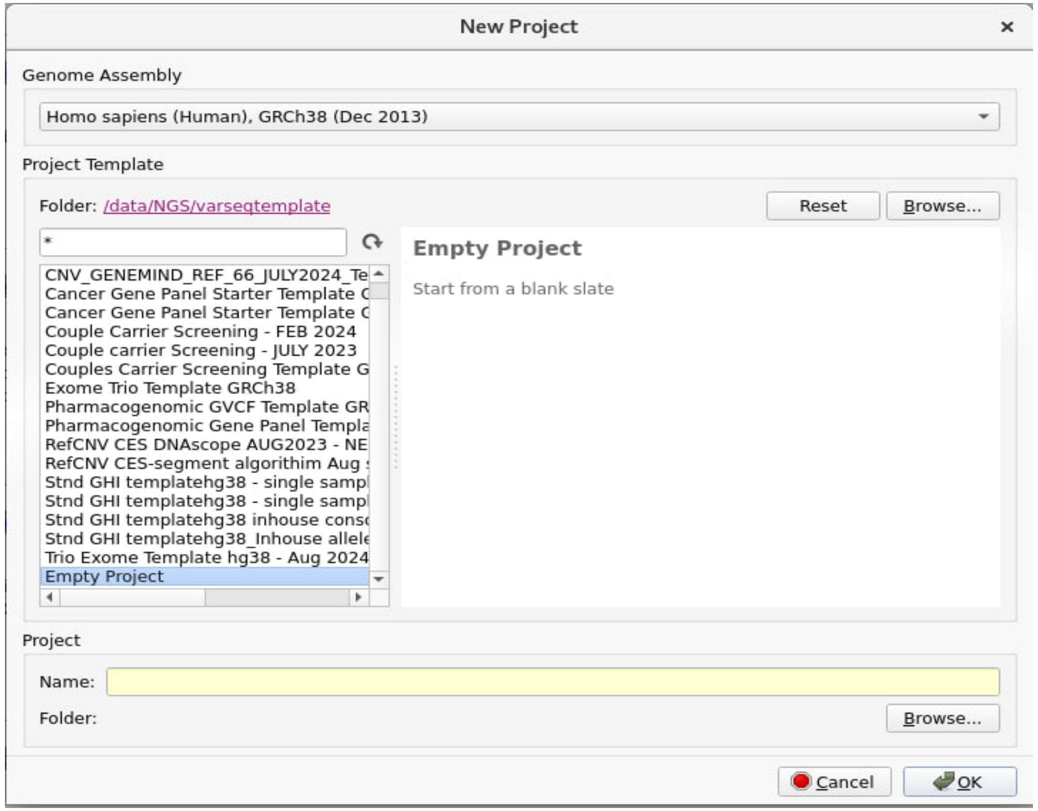
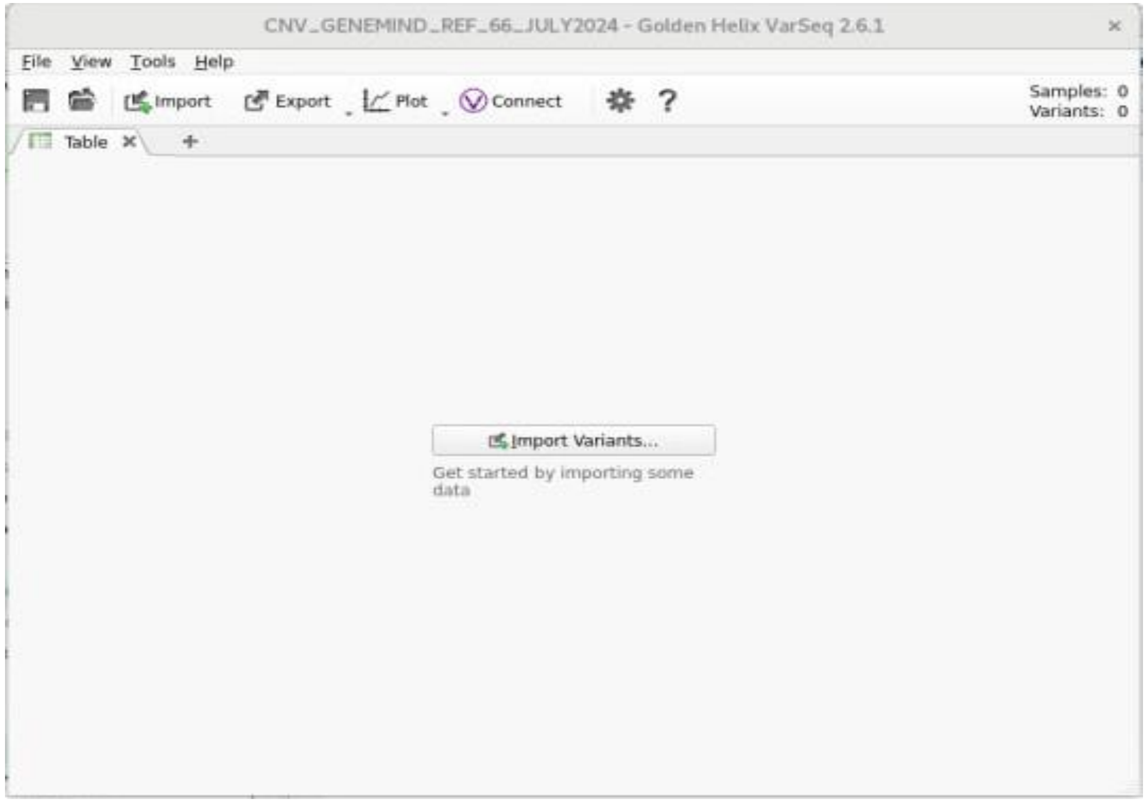
CNV OPTIMISATION - SOP

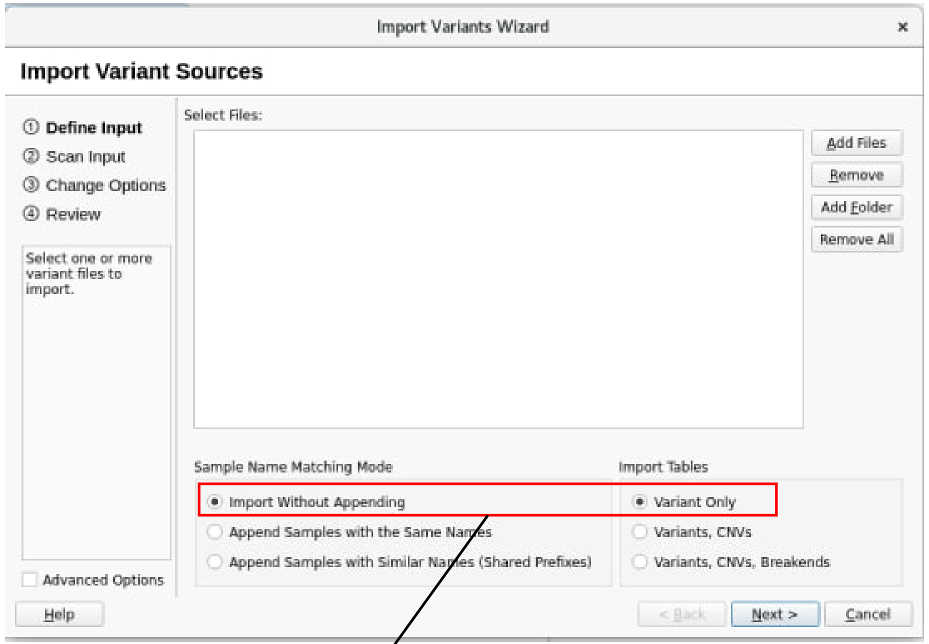
STARTING WITH REFERENCE BAM FILE CREATION

Create a folder with VCF and BAM. Remove the VCF that has a true CNV call.

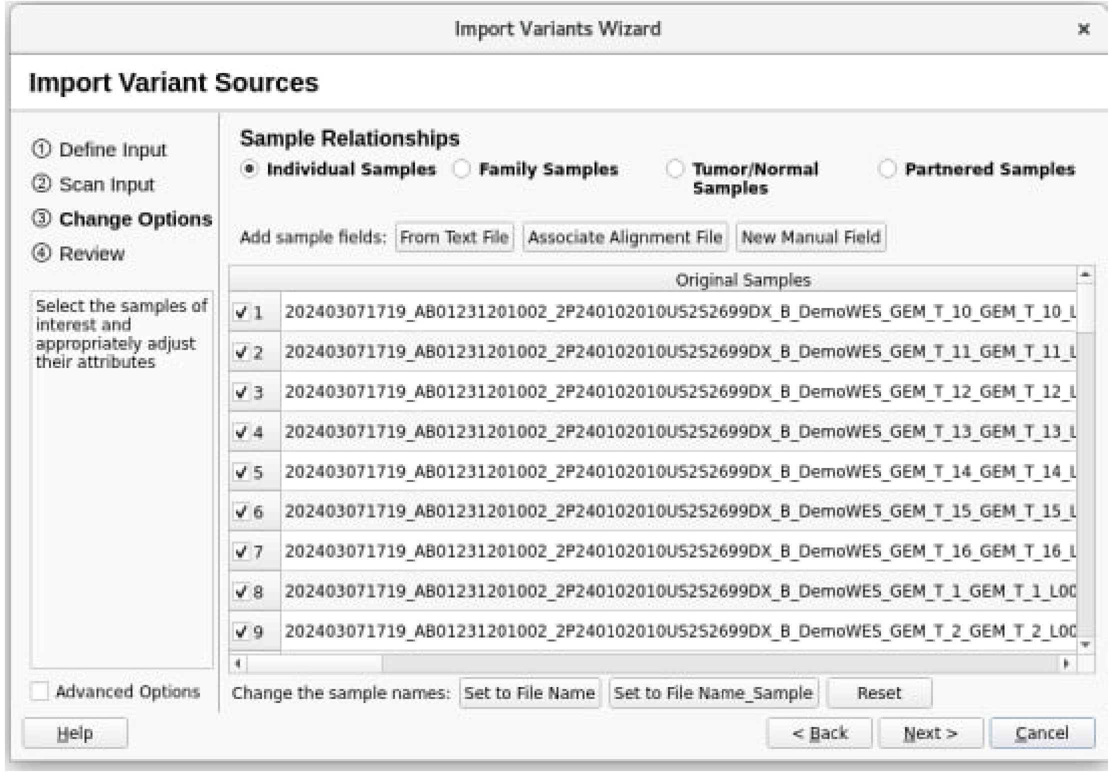


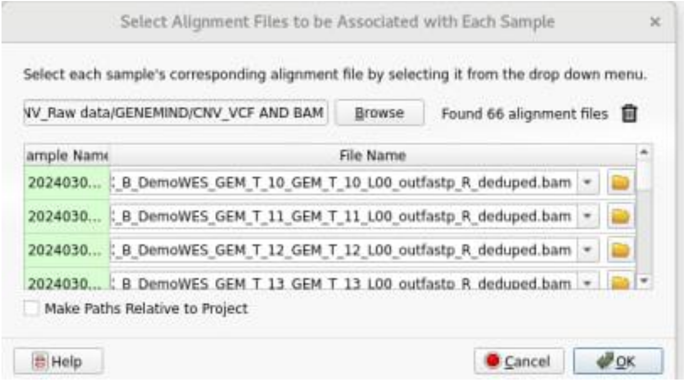


Start with empty project, create a name and locate it in the folder that contains VCF and BAM for CNV

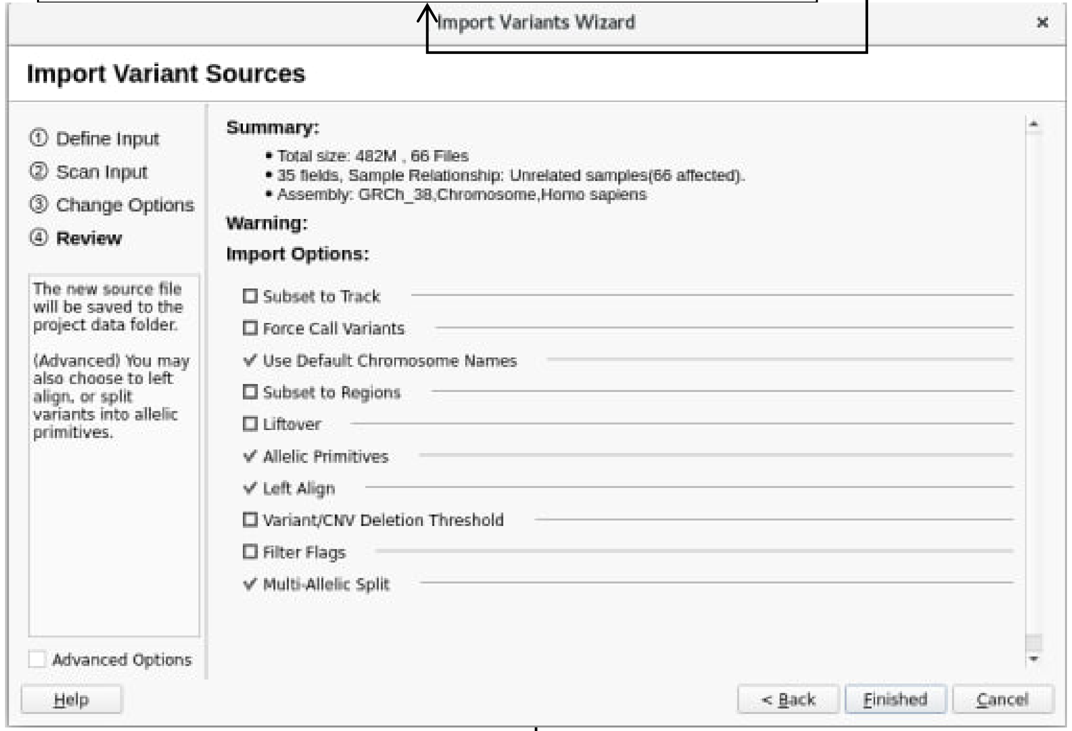


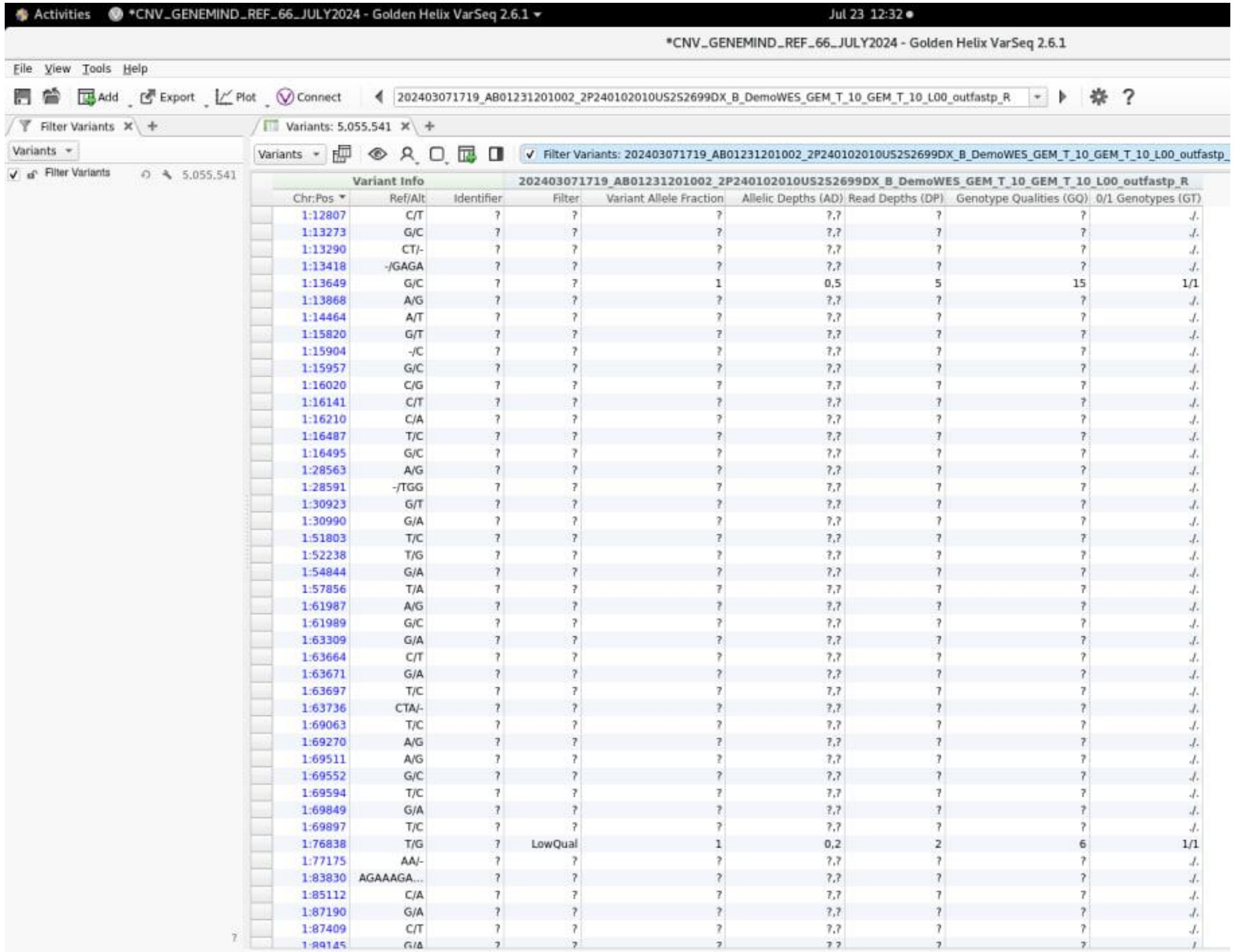
DON'T CHANGE IT

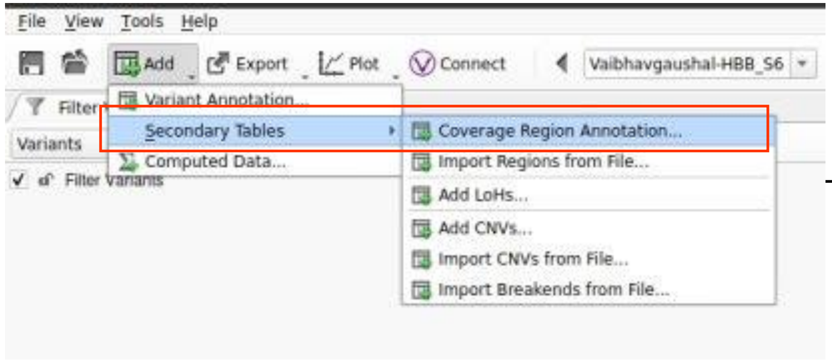


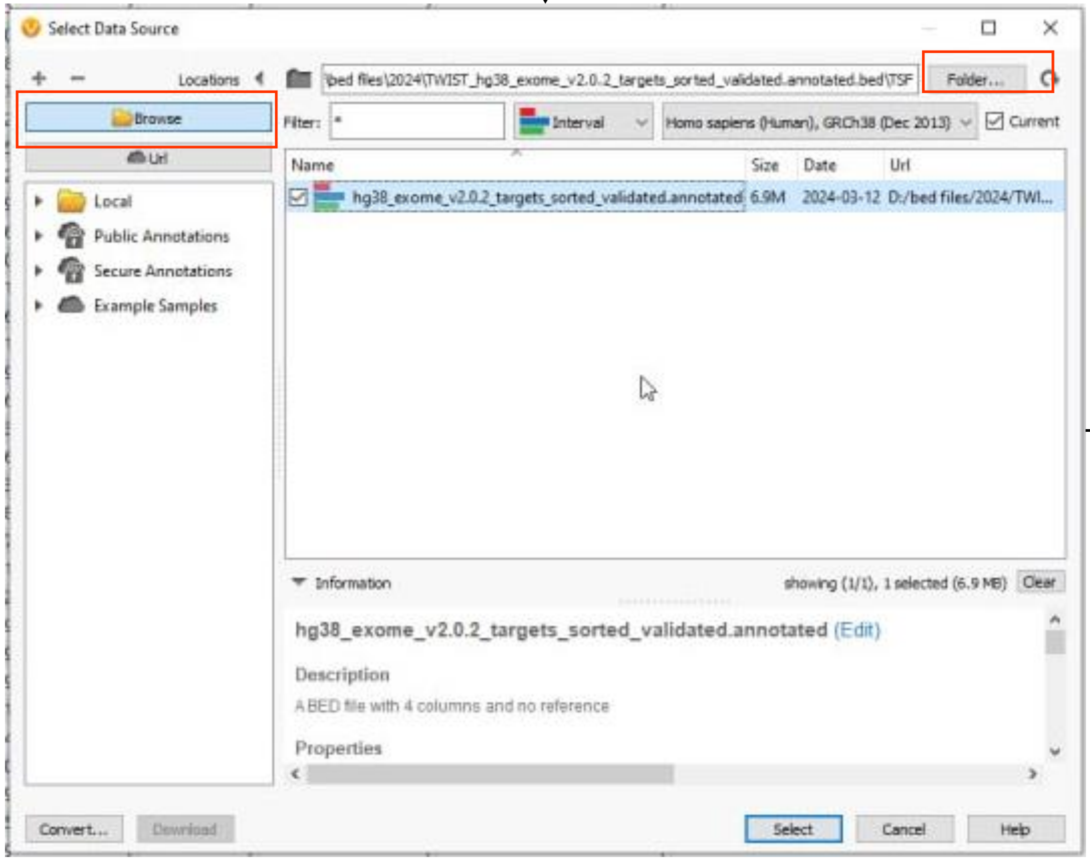


Check whether the BAM file is aligned to its respective VCF.

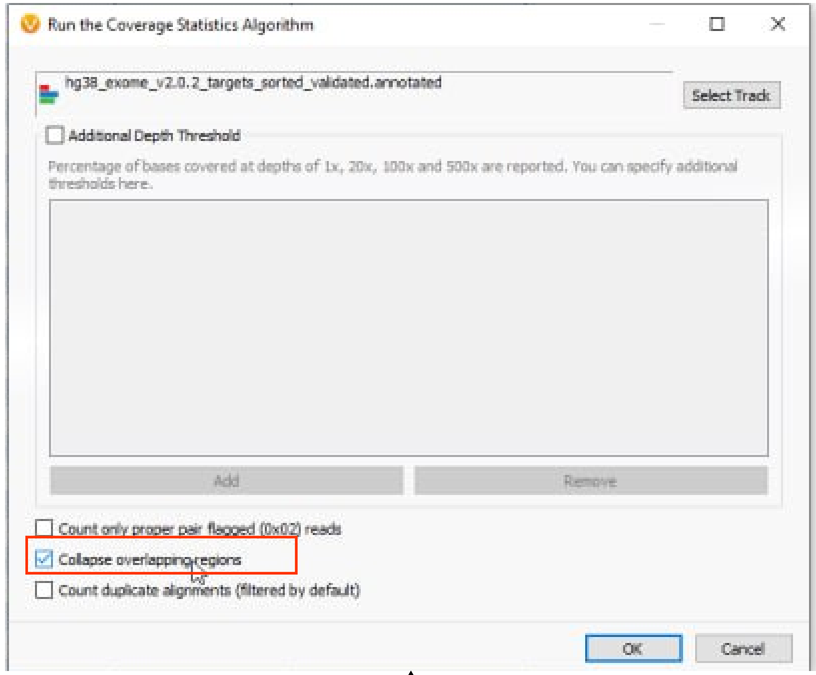




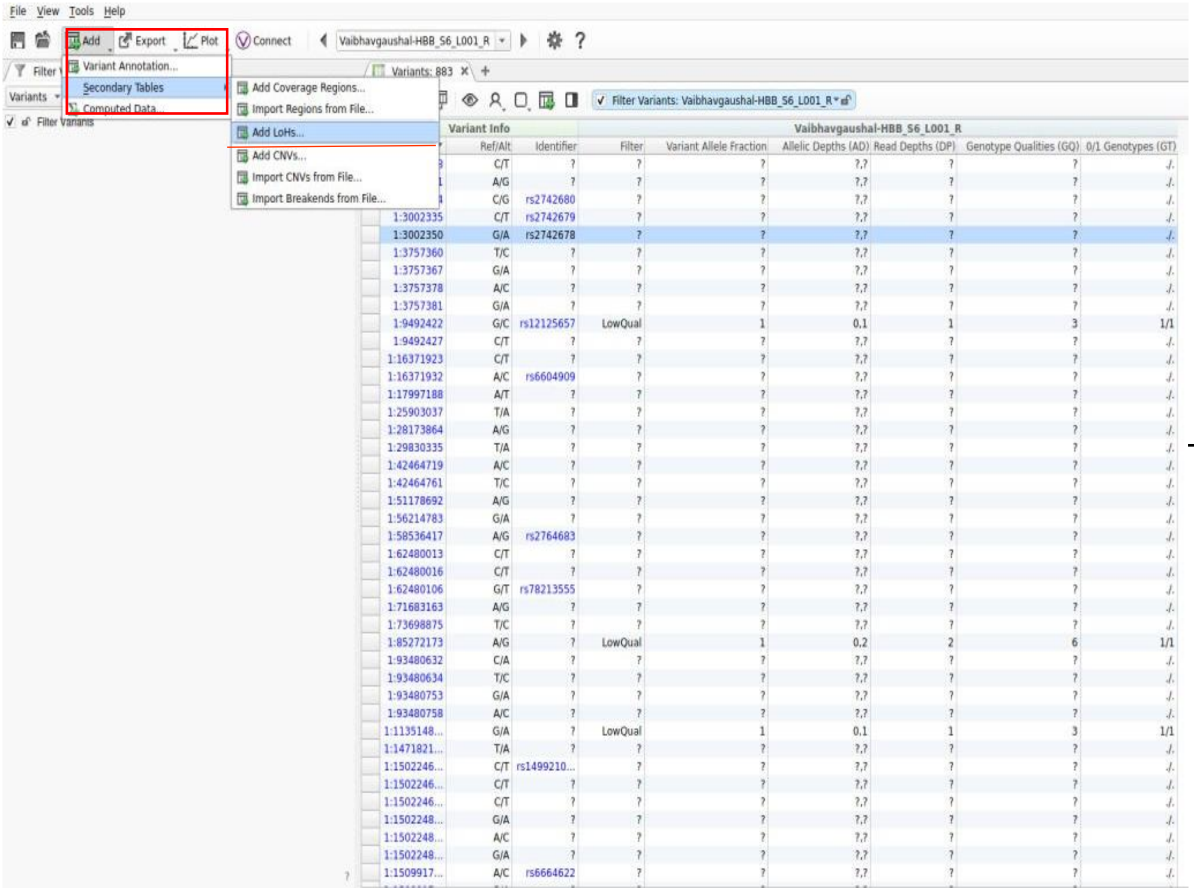


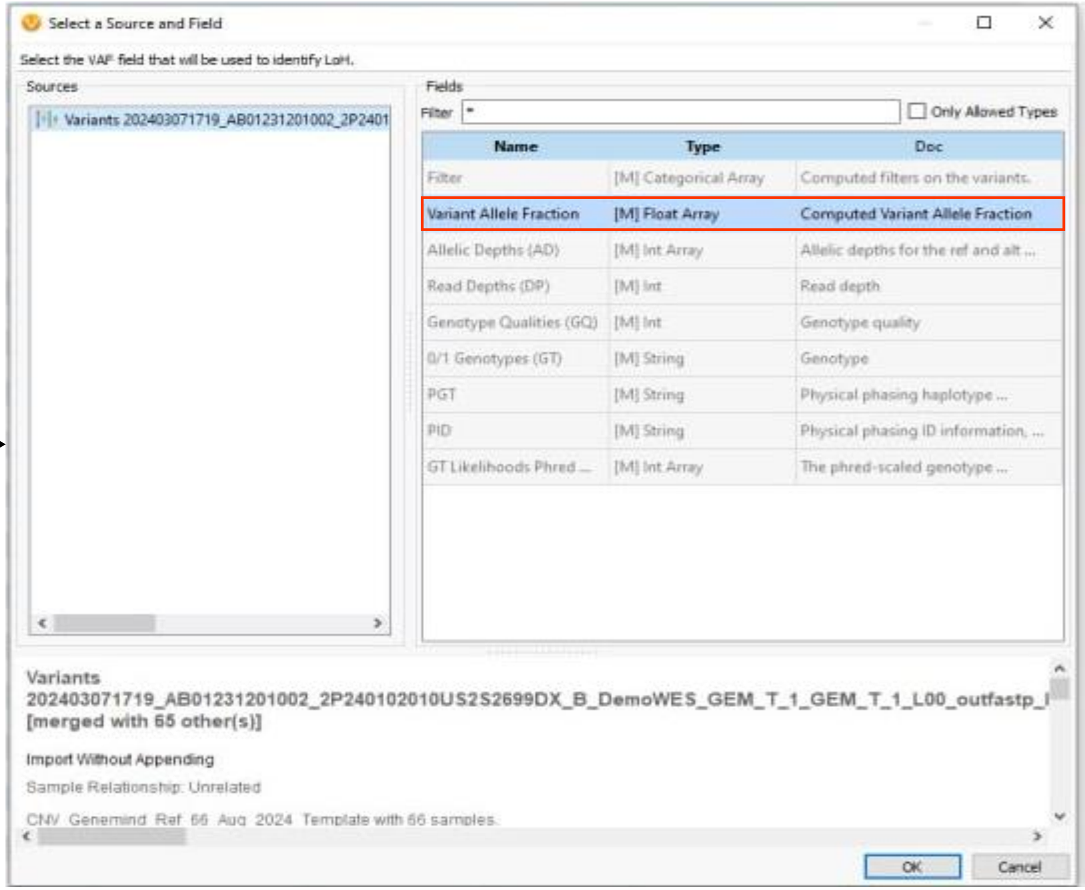


Browse to the folder that has the TSF file.



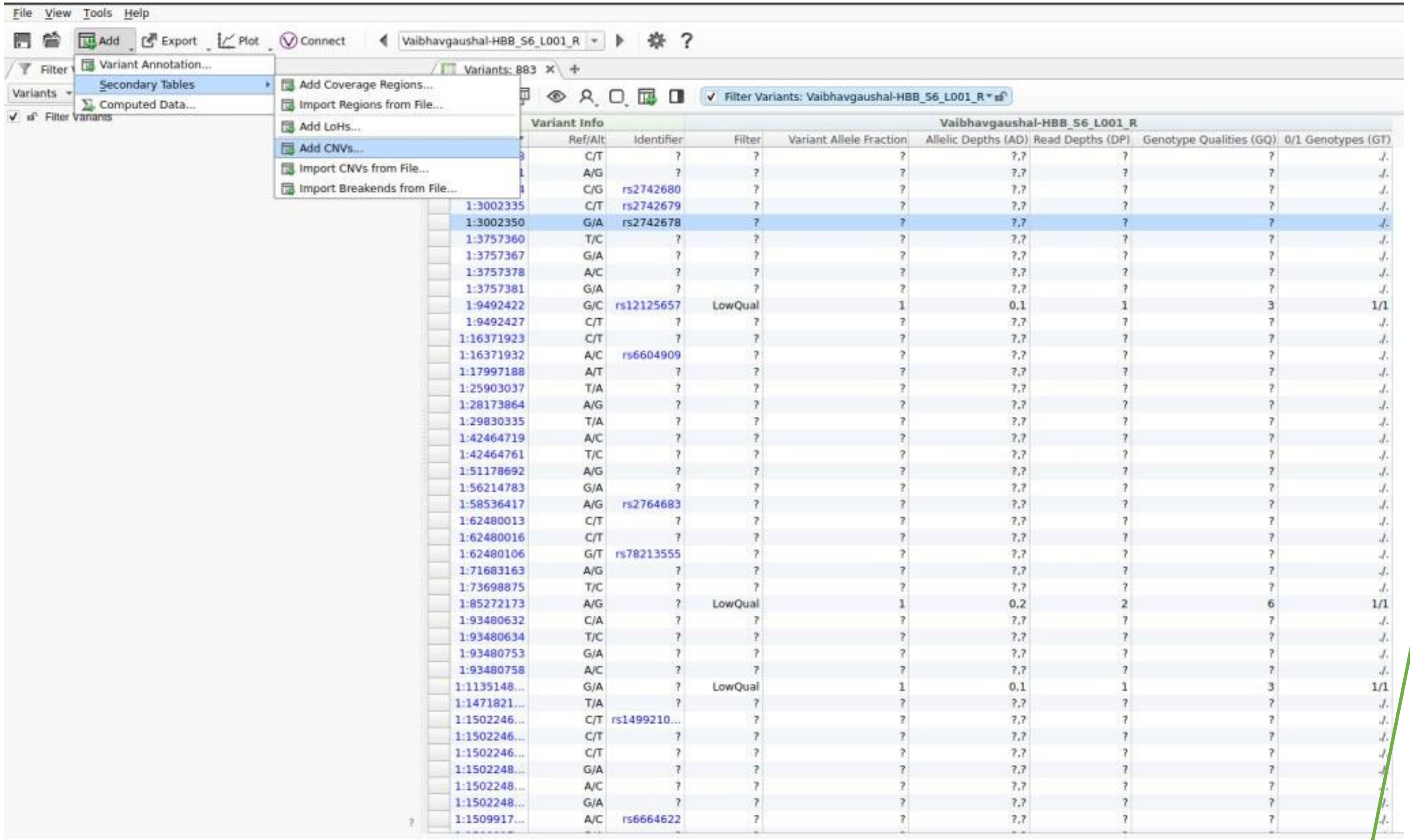
Click on collapse overlapping regions

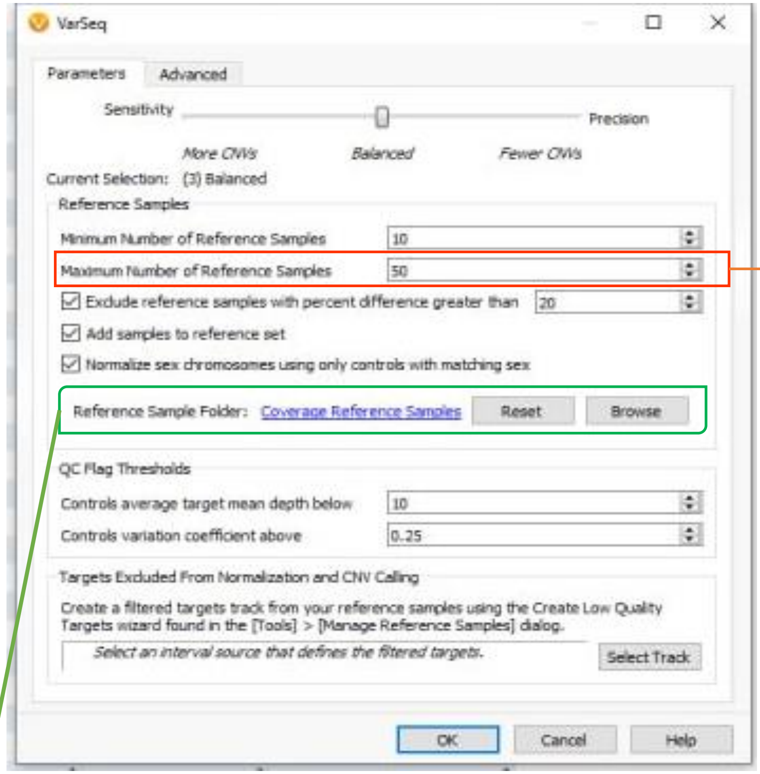




No changes, keep the default value.

ADD LOHs will run only if the filter chain contains the Variant Allele Fraction





Change the Max.no of reference samples to 100 (incase if your reference samples is more than 50).

Note: Only one coverage reference samples folder should be there.

Orelse, don't change any parameters.

Variants: 5,055,541 | LoHs: 15,603 | Coverage Regions: 283,942 | CNVs: 521 | Samples: 66

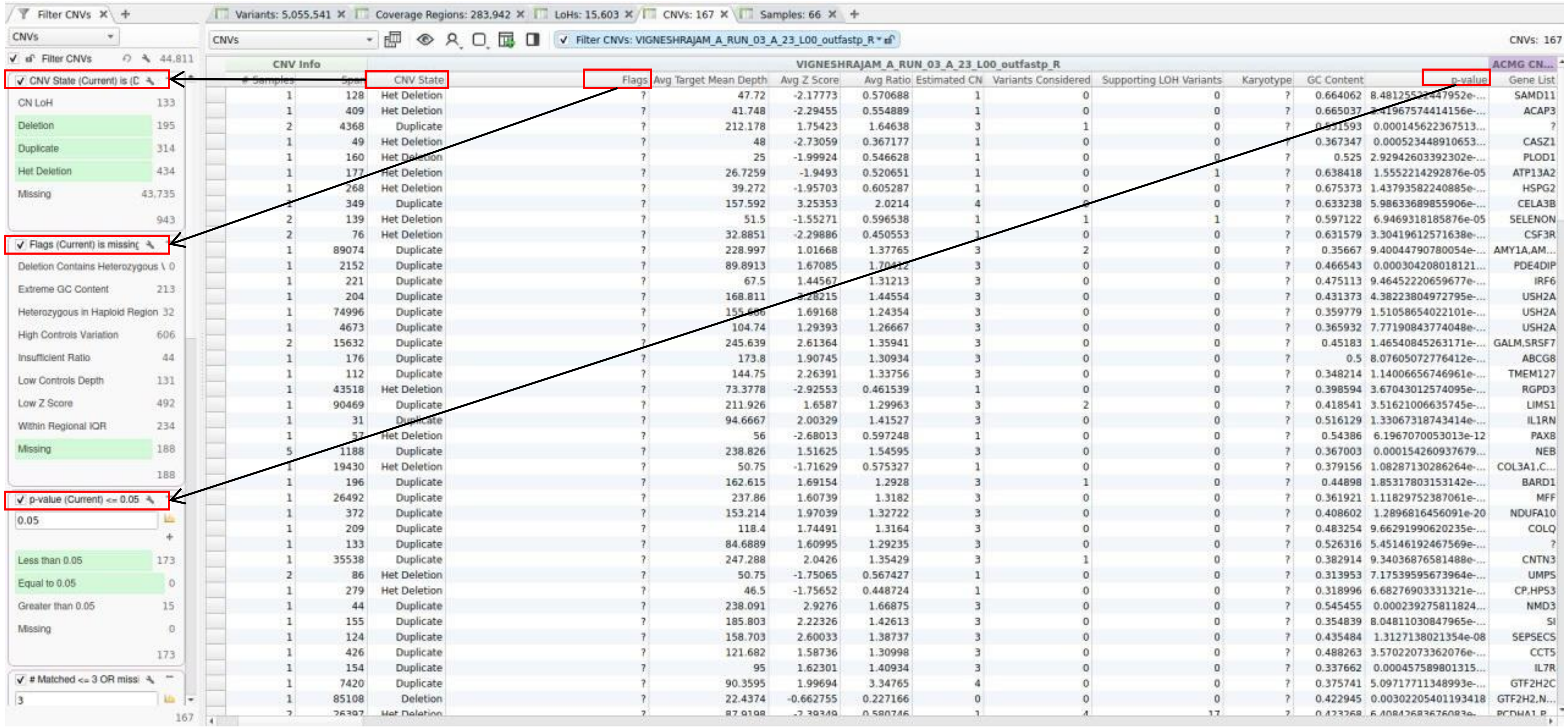
Samples

Samples:

Coverage Statistics hg38_exome_v2.0.2_targets_sorted_validated.annotated								Copy Number Variants						
Sample Mean Reverse ...	Sample Mean Filtered Depth	Sample % 1x	Sample % 20x	Sample % 100x	Sample % 500x	Sample Errors	Sample Flags	Inferred Gender	# CNV Events	# Flagged CNV Events	# Unflagged CNV Events	# Hom Deletions	# Het Deletions	# Duplications
35.3149	1.67922	99.132	97.9447	13.619	0.00990442	?	?	Male	1752	912	840	91	235	129
35.2163	1.77283	99.0737	98.1753	9.83782	0.0101348	?	?	Female	1400	679	721	87	160	105
33.851	1.5525	99.0529	98	9.09599	0.00771019	?	?	Female	1500	737	763	91	167	112
41.793	1.9841	99.0528	98.3599	26.481	0.0109193	?	?	Female	1503	683	820	79	173	110
40.5406	1.86458	99.1834	98.3851	24.7453	0.0113417	?	?	Male	1520	736	784	76	193	110
34.4396	1.62866	99.0584	98.0584	9.55871	0.00570787	?	?	Female	1364	656	708	74	183	100
34.9908	1.66418	99.1953	98.0632	11.3464	0.00744136	?	?	Male	1571	761	810	82	217	114
34.2583	1.43485	99.189	97.8825	11.9182	0.00355747	?	?	Male	1818	831	987	78	226	135
51.5125	2.24522	99.105	98.4468	50.2152	0.0144575	?	?	Male	1182	669	513	90	129	83
41.5718	1.85394	99.1167	98.1901	29.618	0.0107931	?	?	Male	1288	719	569	101	164	89
47.9647	2.21702	99.0793	98.299	43.4938	0.010442	?	?	Male	1333	782	551	117	225	80
58.4068	2.55912	99.201	98.5872	60.2147	0.0217809	?	?	Male	1417	658	759	69	184	100
46.3105	2.033	99.1077	98.3538	39.8111	0.00934757	?	?	Male	1572	817	755	94	152	112
55.2905	2.56881	98.9441	98.3663	56.9045	0.013259	?	?	Female	1081	671	410	120	131	68
44.171	2.01153	99.0926	98.2655	34.6983	0.00771282	?	?	Male	1084	689	395	100	177	70
76.529	3.80884	98.8825	98.4077	83.7633	0.0308268	?	?	Male	1038	656	382	222	398	31
80.8855	4.47105	98.8408	98.3024	83.3655	0.035536	?	?	Male	1237	698	539	219	683	25
72.2702	3.76716	98.715	98.2476	79.3735	0.0178531	?	?	Female	1032	670	362	280	445	22
54.5259	2.9597	98.7278	97.9262	53.6292	0.00779236	?	?	Male	1643	1050	593	331	939	31
80.4617	4.05761	98.9317	98.4655	86.3529	0.0352702	?	?	Male	869	588	281	197	285	25
78.3613	3.97841	98.8877	98.3403	84.9103	0.0356432	?	?	Female	848	541	307	165	362	24
76.3105	3.62403	98.9559	98.464	84.8187	0.0295679	?	?	Male	930	613	317	160	262	38
50.6683	3.16001	98.5879	97.8096	47.1835	0.00539787	?	?	Female	3914	1946	1968	522	2164	118
66.1304	3.28323	98.8647	98.3437	74.0182	0.0191422	?	?	Male	932	613	319	254	346	23
63.8663	3.16748	98.7773	98.2254	71.0504	0.0171483	?	?	Female	936	658	278	266	378	18
81.5086	4.27135	98.8619	98.3452	84.8336	0.0376617	?	?	Male	995	660	335	213	501	21
75.7372	3.69943	98.7951	98.2282	81.6243	0.0254233	?	?	Female	922	691	231	249	423	15
107.853	5.44413	98.7992	98.4075	94.9916	0.142368	?	?	Female	822	562	260	232	320	12
82.7679	4.25163	98.9051	98.4646	87.552	0.0430954	?	?	Male	1017	649	368	236	351	31
88.0069	3.95509	98.8864	98.3416	80.8051	0.0662641	?	?	Female	1333	907	426	223	869	17
66.9024	4.28049	98.6656	98.1609	73.5112	0.0144849	?	?	Female	3226	1850	1376	511	1706	95
88.3255	4.48773	98.9334	98.4491	88.3705	0.0449112	?	?	Male	823	584	239	207	378	16
59.716	2.78449	98.796	98.0778	64.2076	0.0106423	?	?	Female	1061	700	361	244	479	22
84.823	4.0765	98.8563	98.3585	88.8274	0.041192	?	?	Female	906	585	321	204	325	20
99.737	4.90884	98.9048	98.4558	91.8447	0.0778767	?	?	Male	772	579	193	224	342	13
68.0092	3.09983	98.9298	98.3333	75.0599	0.0230481	?	?	Male	874	644	230	191	360	23
87.7094	5.02534	98.8581	98.4259	88.7435	0.0412468	?	?	Male	1990	1491	499	388	1236	28
72.2978	3.83474	98.7419	98.2303	70.0424	0.0594444	?	?	Male	1298	874	424	344	555	27
64.3078	3.2146	98.8541	98.1796	63.4304	0.020969	?	?	Male	966	713	253	283	412	18

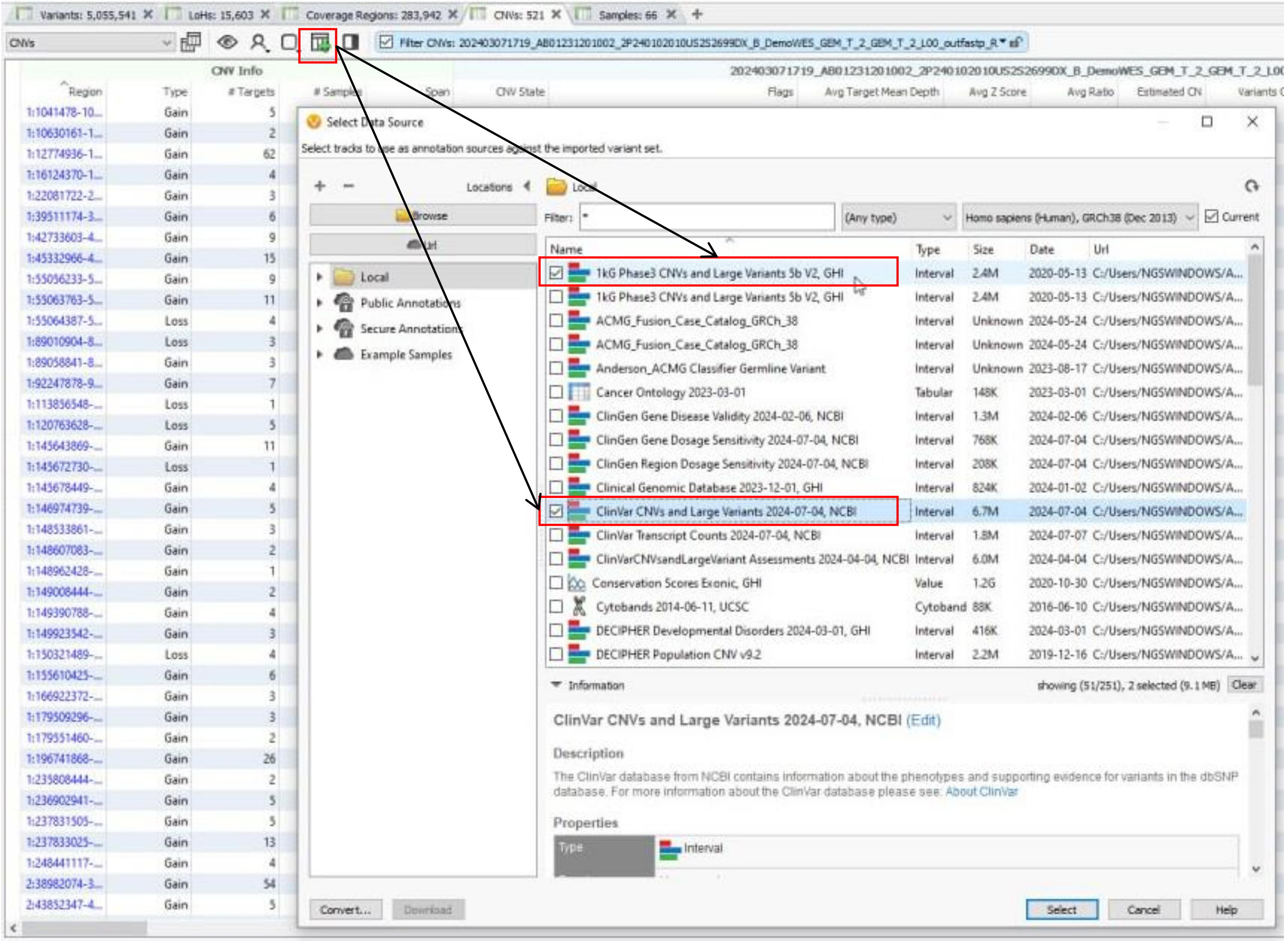
go to the samples table, and check the sample flags, (Check the CNV caller tutorial in Golden Helix web page to know what sample flags to remove)

Building the workflow

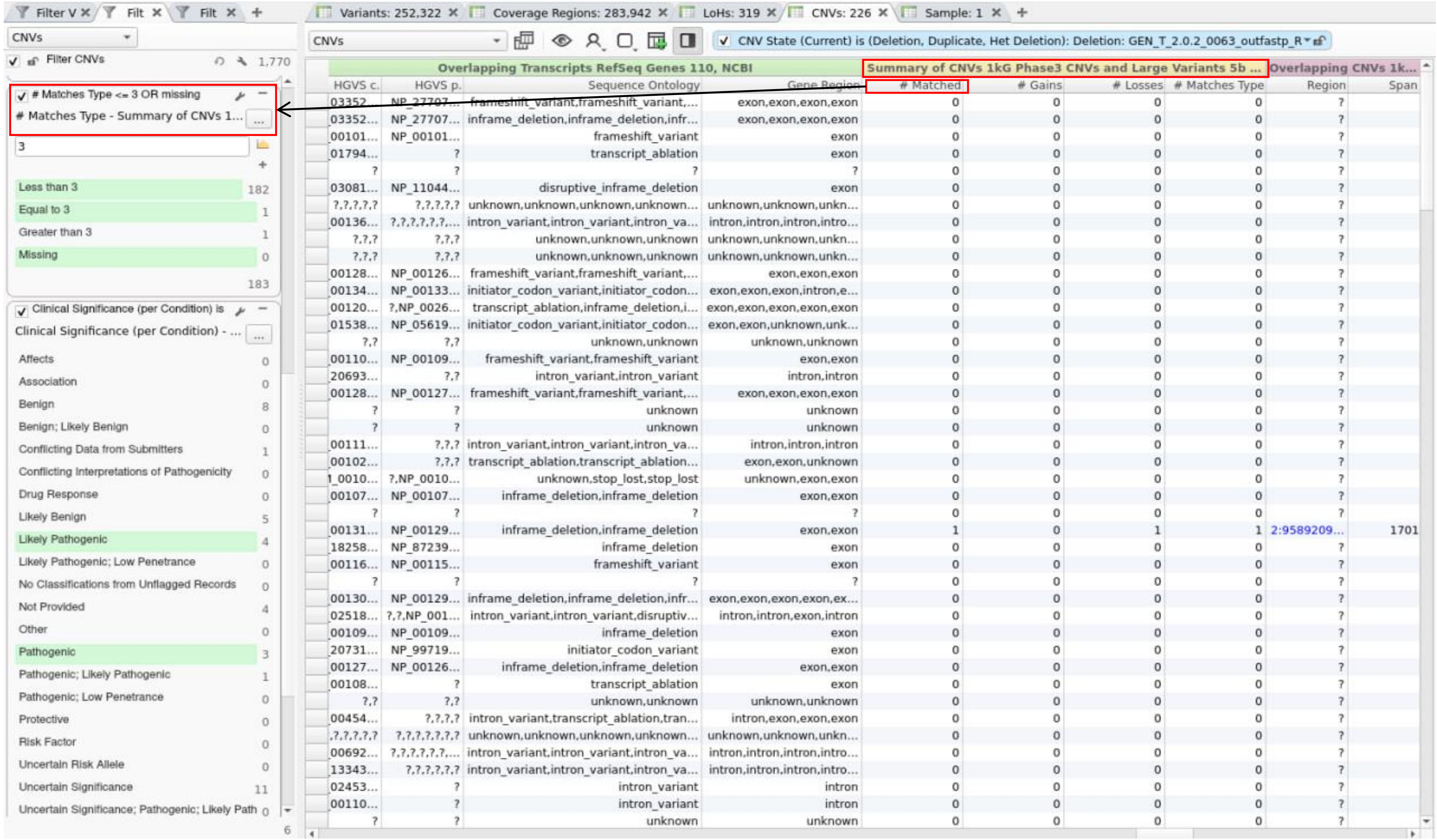


Add the below to filter chain

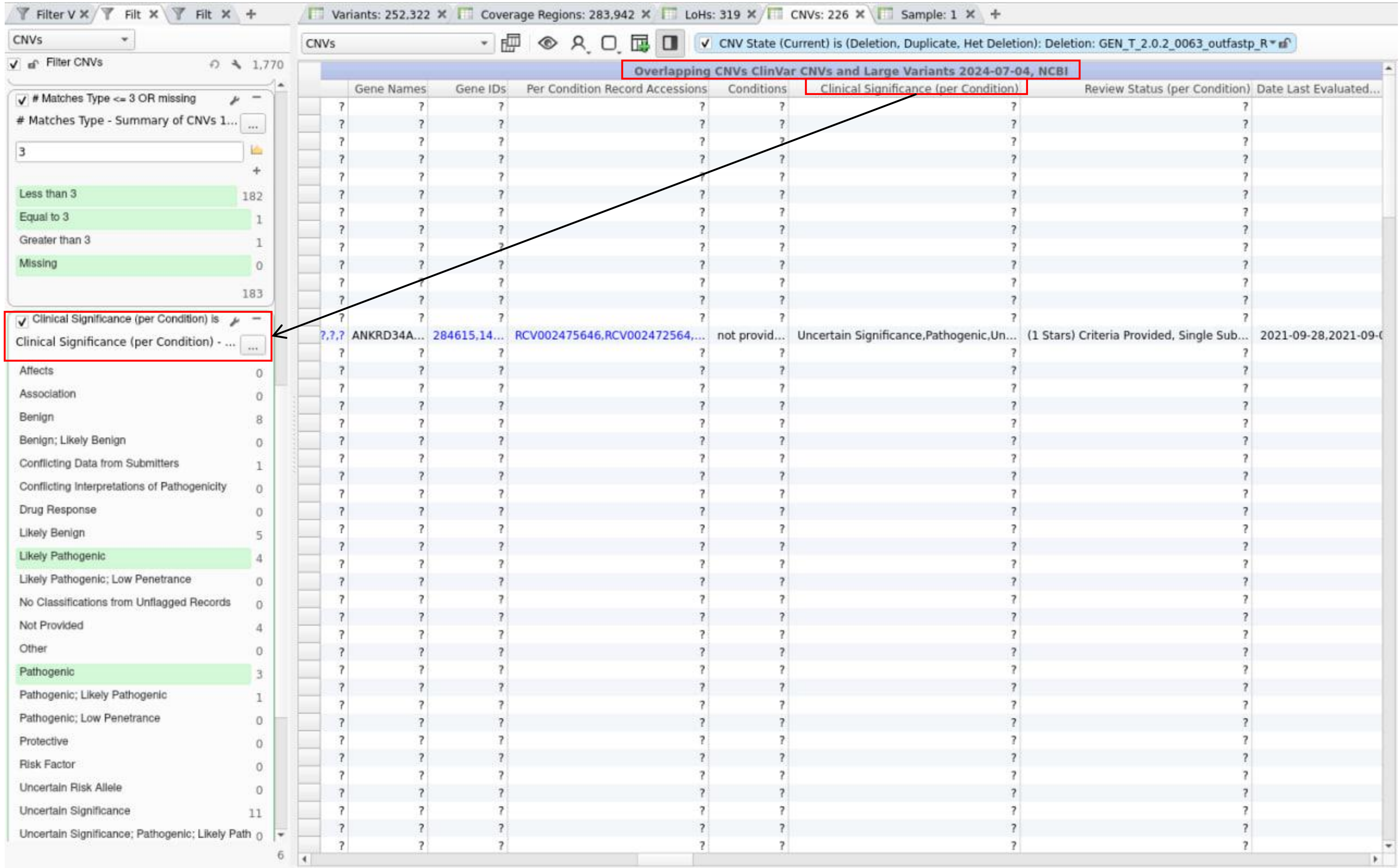
* CNV state
* Filter flags
* p-Value - 0.05



Add 1Kg phase 3 CNV and ClinVar CNV db

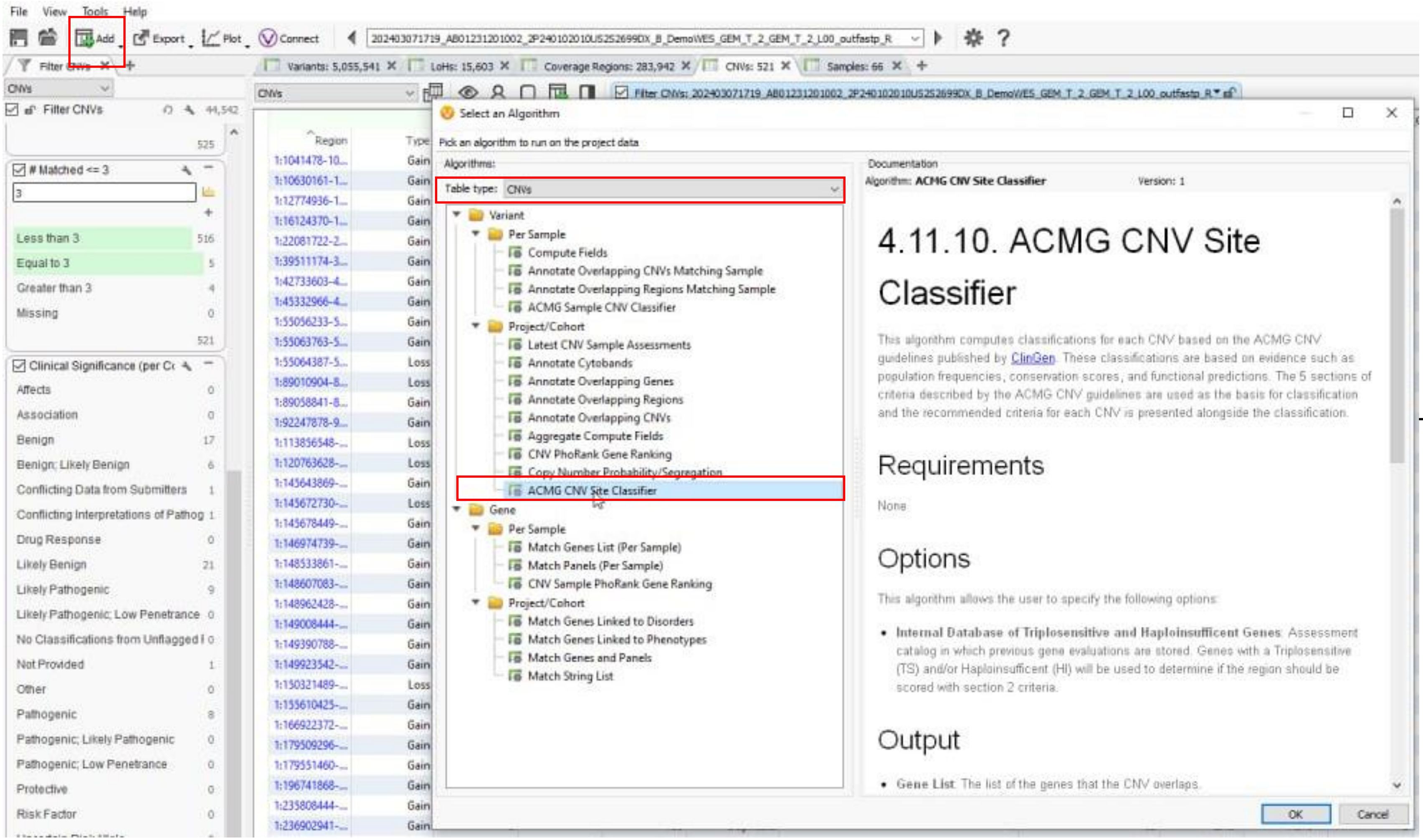


Add Matched to the filter chain - Keep the value 3.

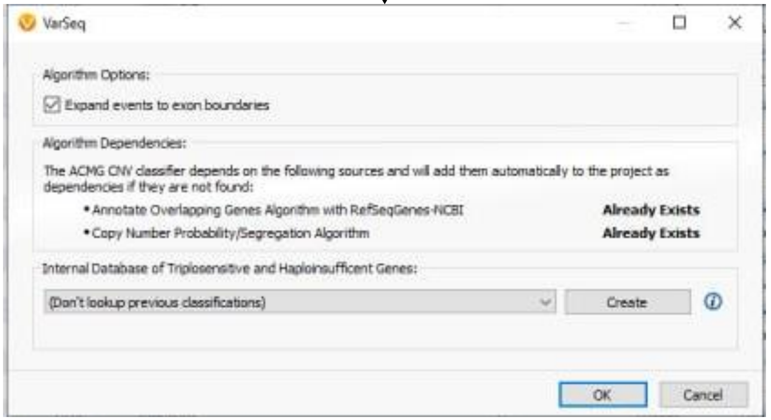


Add Clinical Significance (Per Condition) from ClinVar.

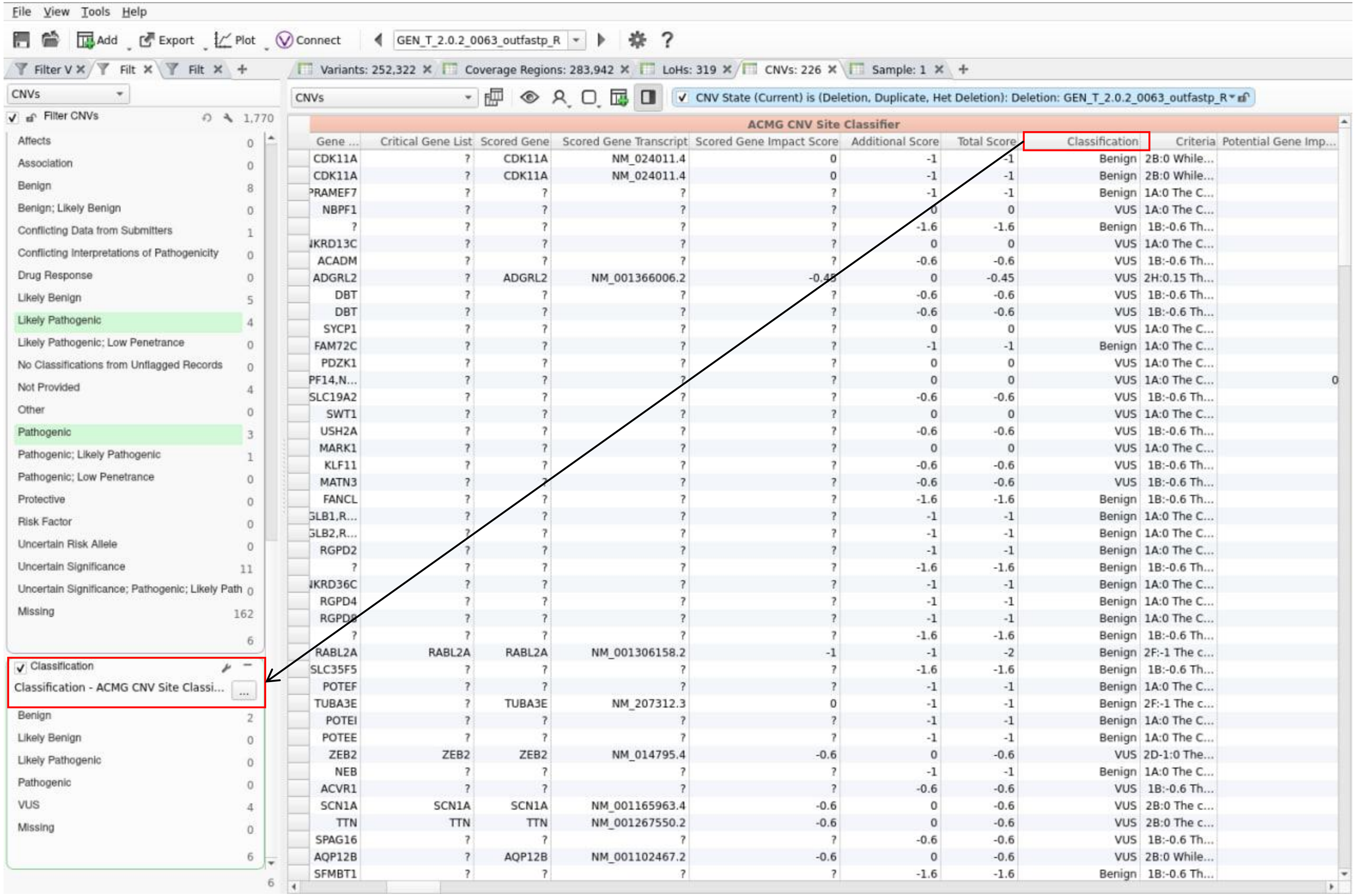
Note: Use the updated Clinvar db.



Click on the ADD --- Computed Data, select **table type** --- CNV, Then choose ACMG CNV Site Classifier.

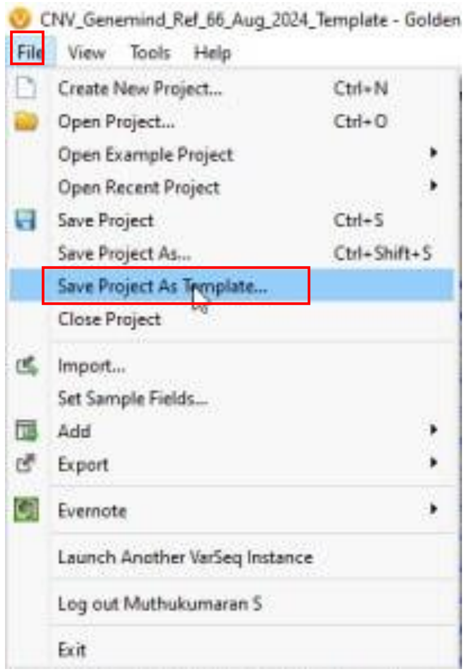


leave the default options, click on

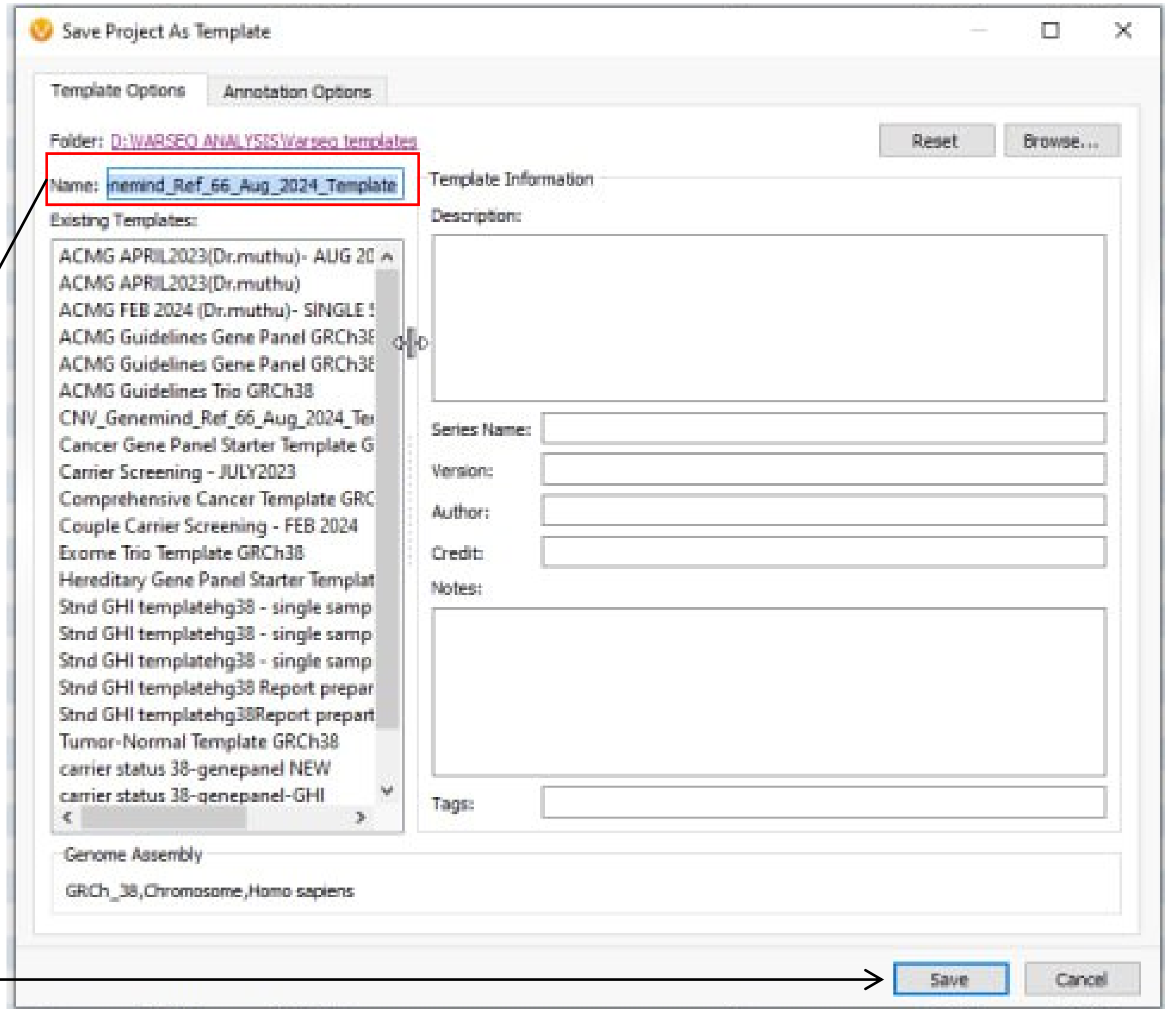


Add **Classification** from **ACMG CNV Site Classifier** to the filter chain.

Template Creation



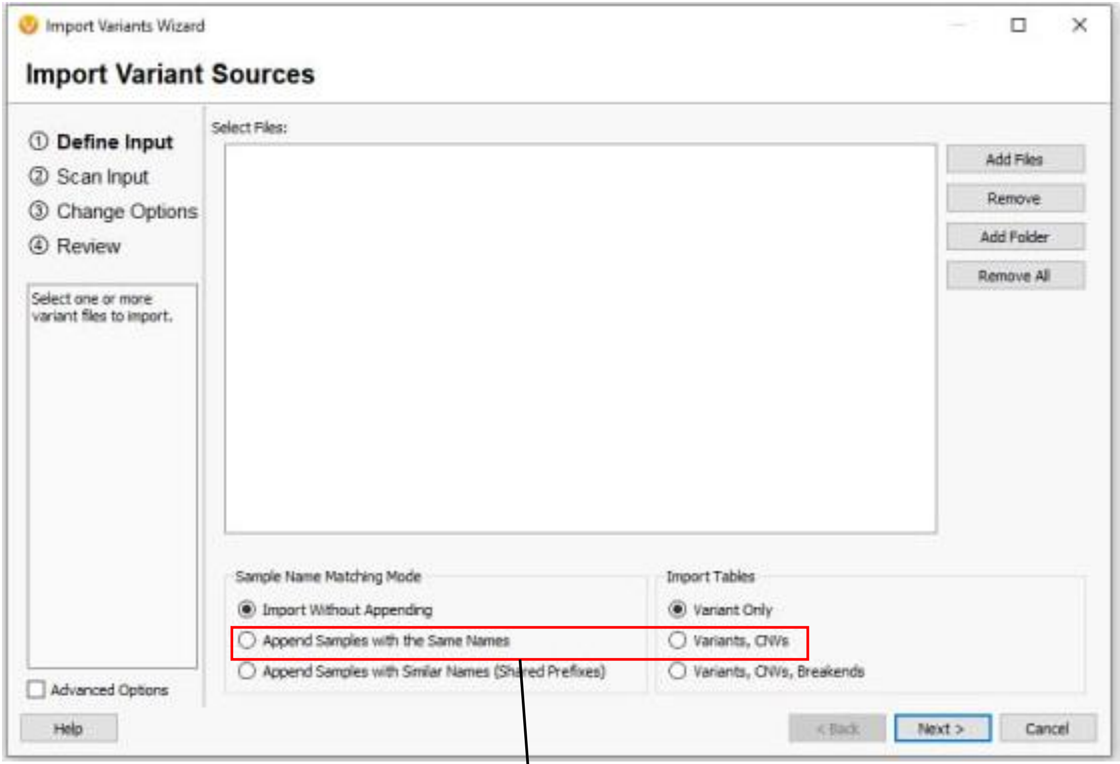
Go to File, Click Save Project as Template

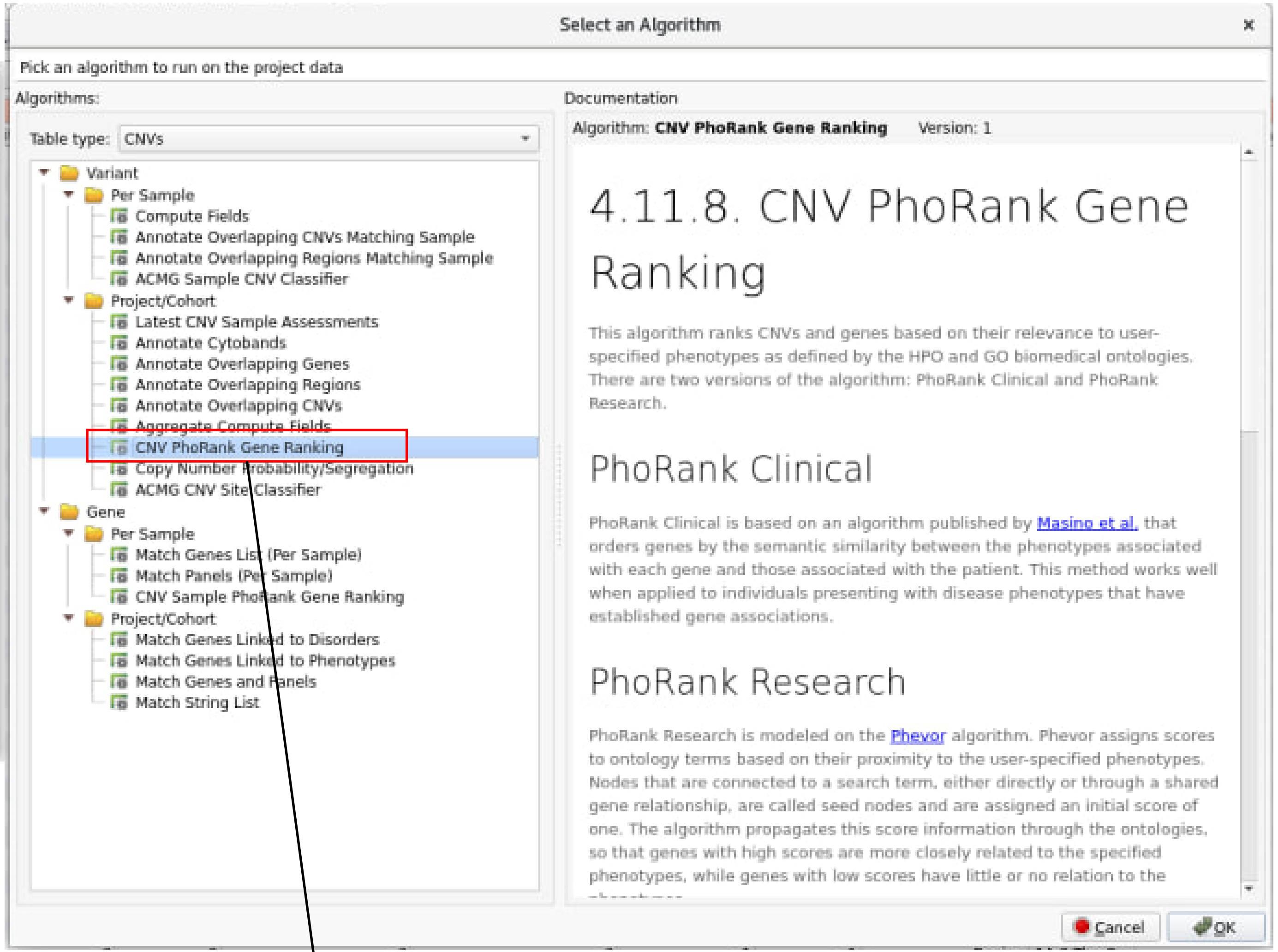


Rename the file and save it.

CNV Analysis

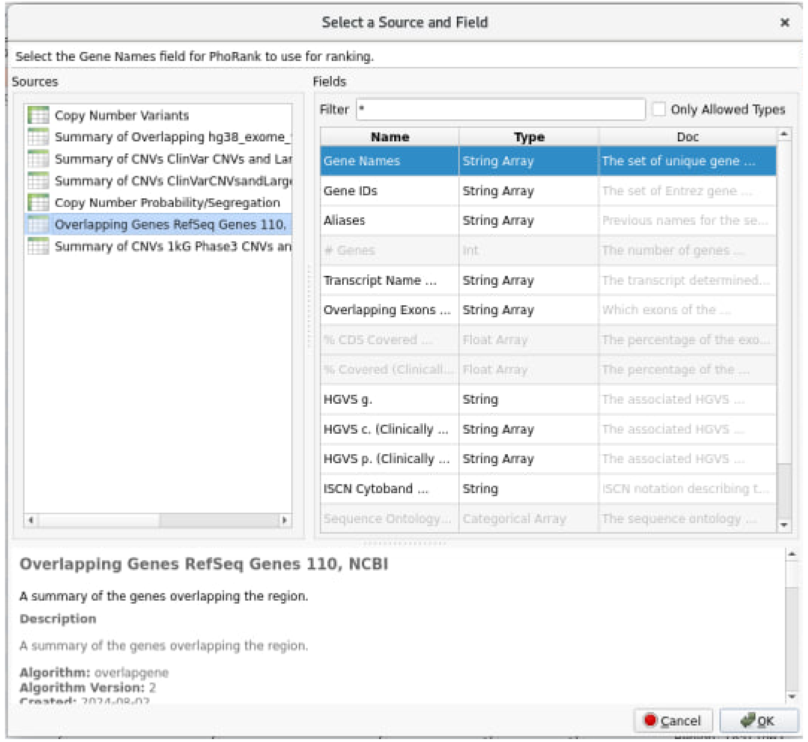
- To do the CNV analysis, Load the VCF file and **choose** --- **Append samples with same names and Variants, CNV** - and BAM in the project template created.
- After loading, check for updates.
- Go to **Add** --- Computed Data --- **Table type** - CNV ---- **CNV PhoRank Gene Ranking**.
- Leave the default option and click OK.
- Go to **Input sources** --- Select PhoRank Research (Ontology Propagation)
- Come back to **Phenotype Selection** --- Type the phenotypes and give ok.
- After loading, add **Gene Rank** to Filter chain and keep it above **ClinVar**.
- **Note: Beware of X linked CNVs - High chance of false call.**

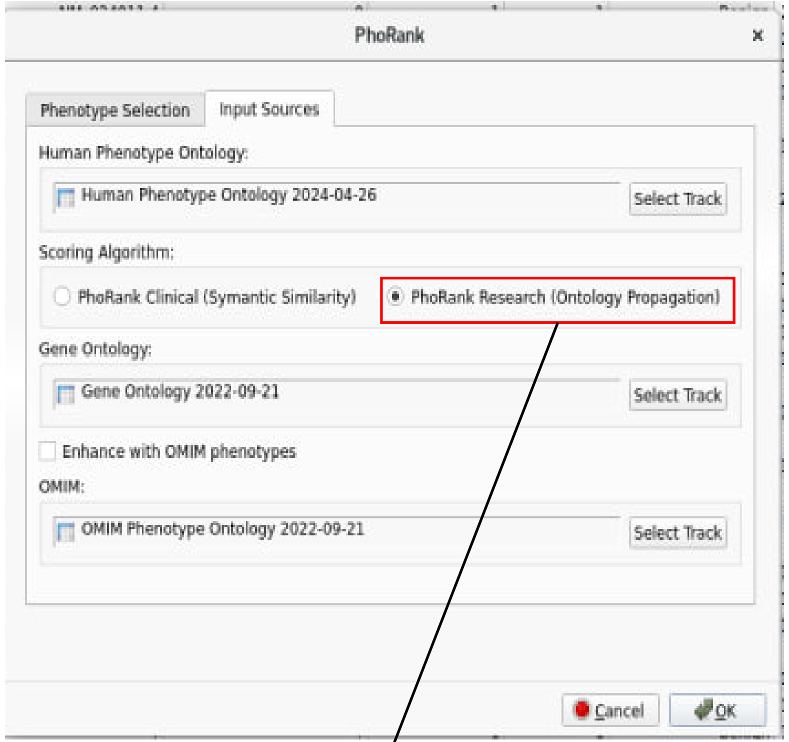


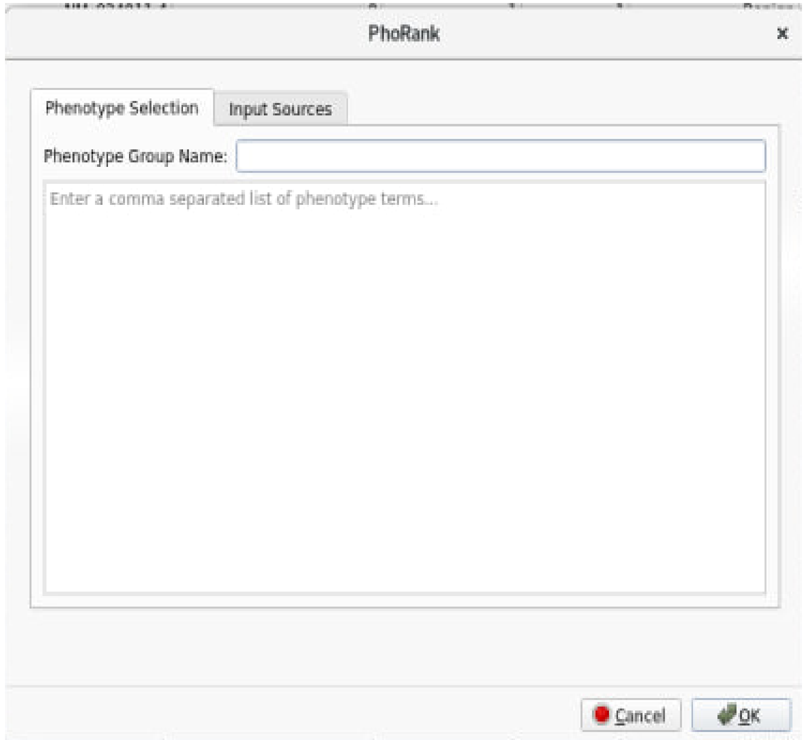


Click on this while loading the VCF file

To add Gene Rank







Leave the default options, Click Ok

Select the second option

Type the phenotypes and give ok